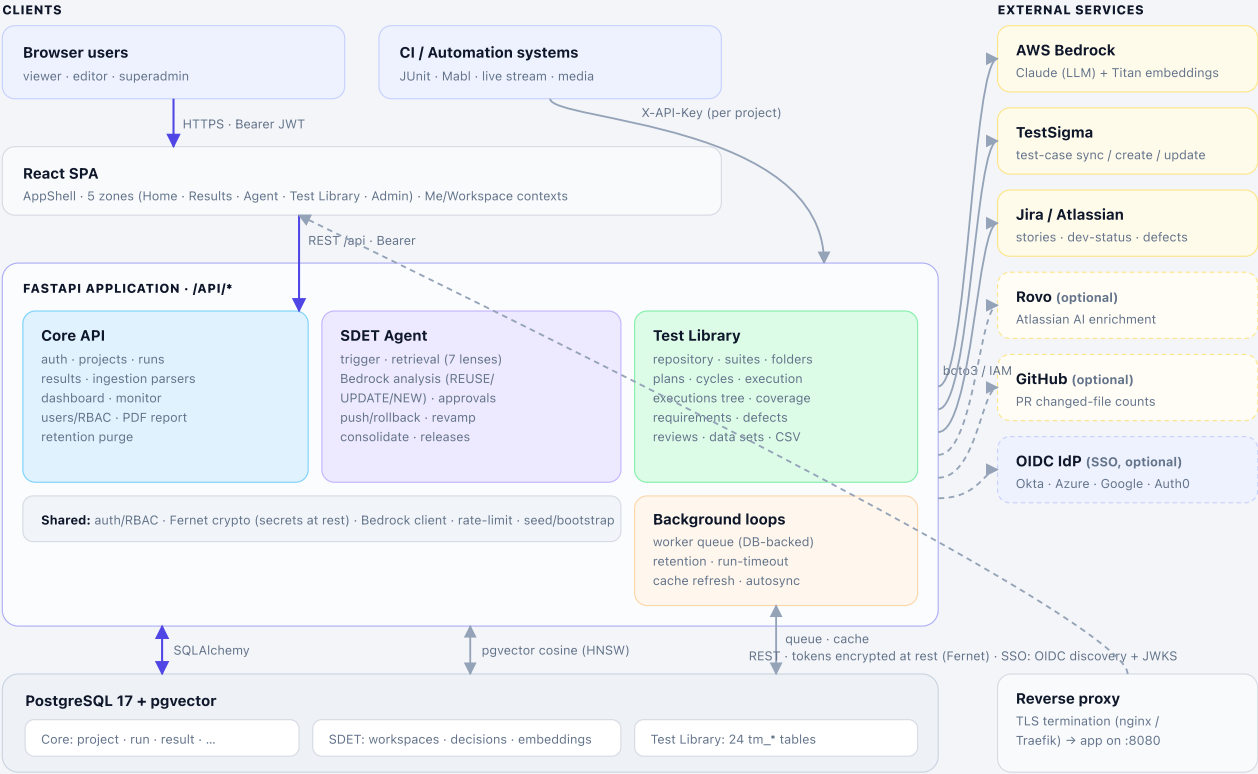
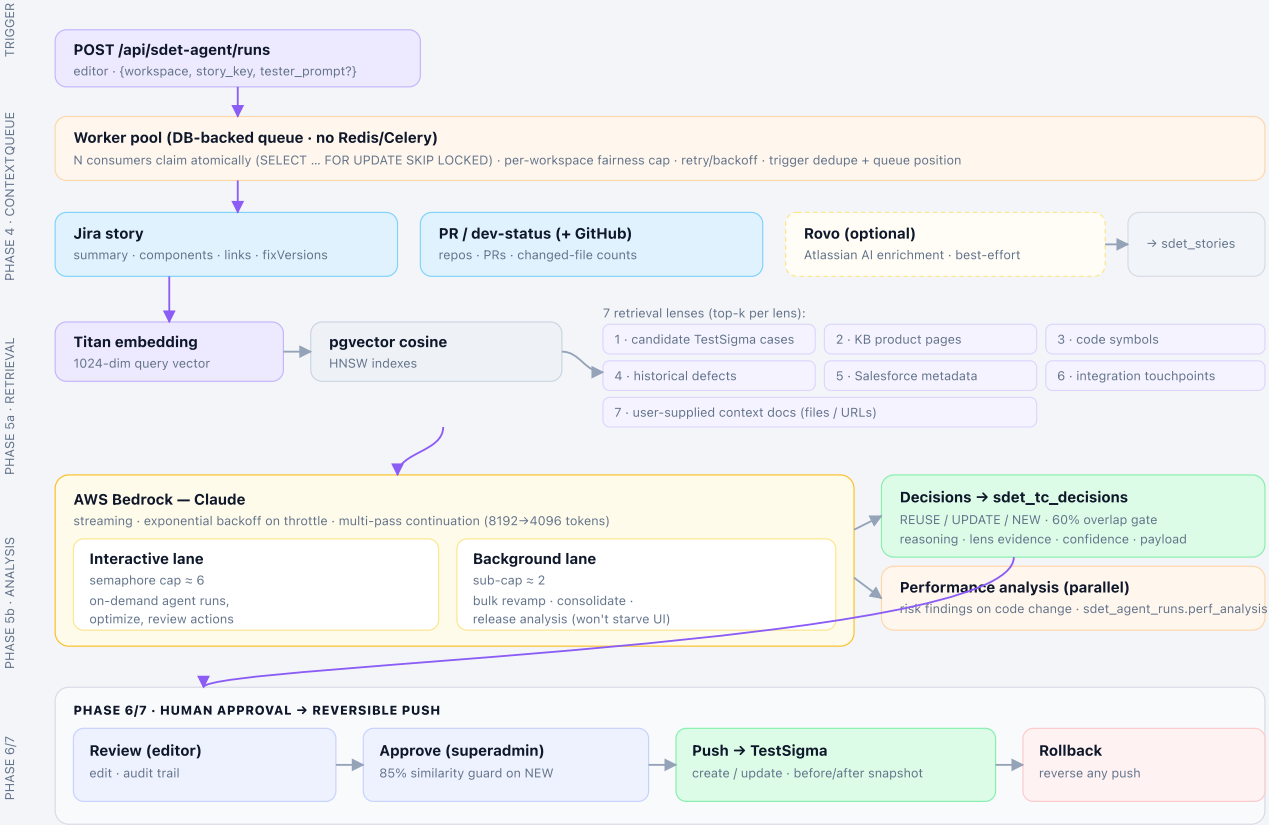




System architecture & data flow

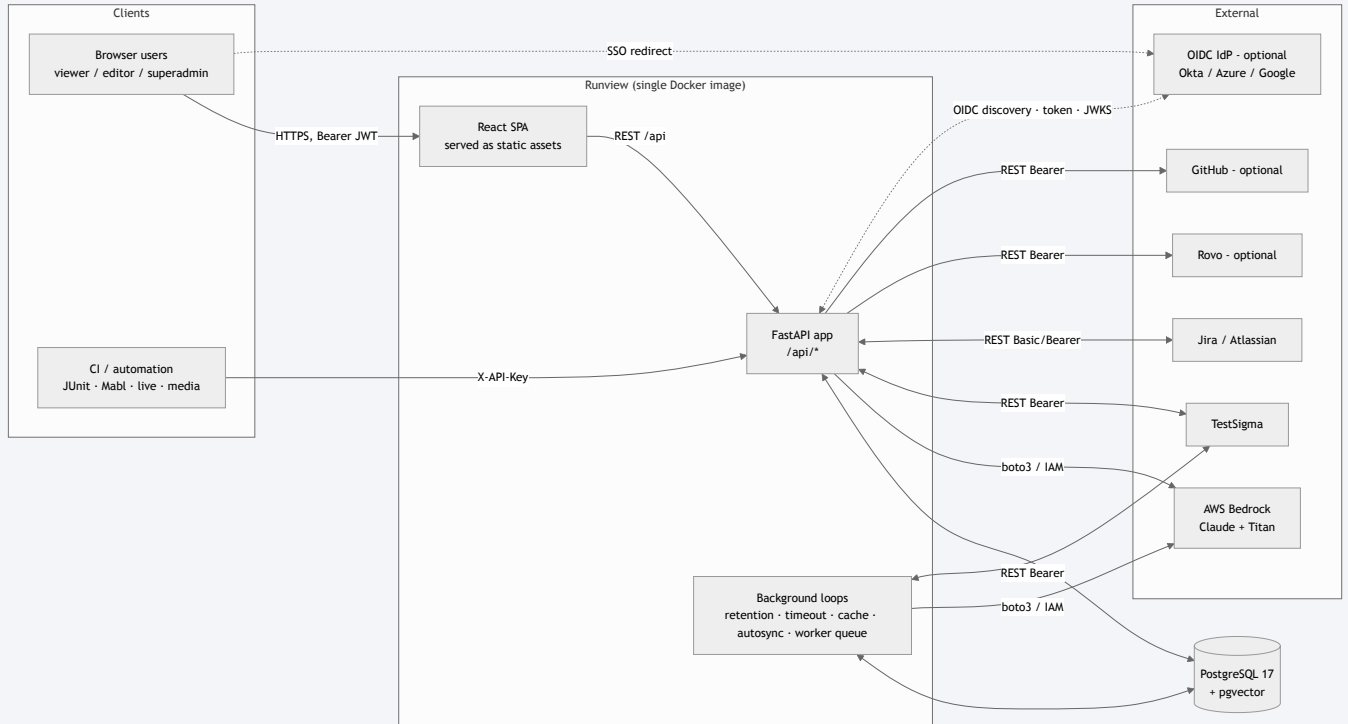


SDET Agent run — worker queue, Bedrock lanes & 7-lens retrieval

Runview — Architecture

This document describes the system topology, components, data flows, and data model. A rendered visual diagram is also available as an interactive artifact.

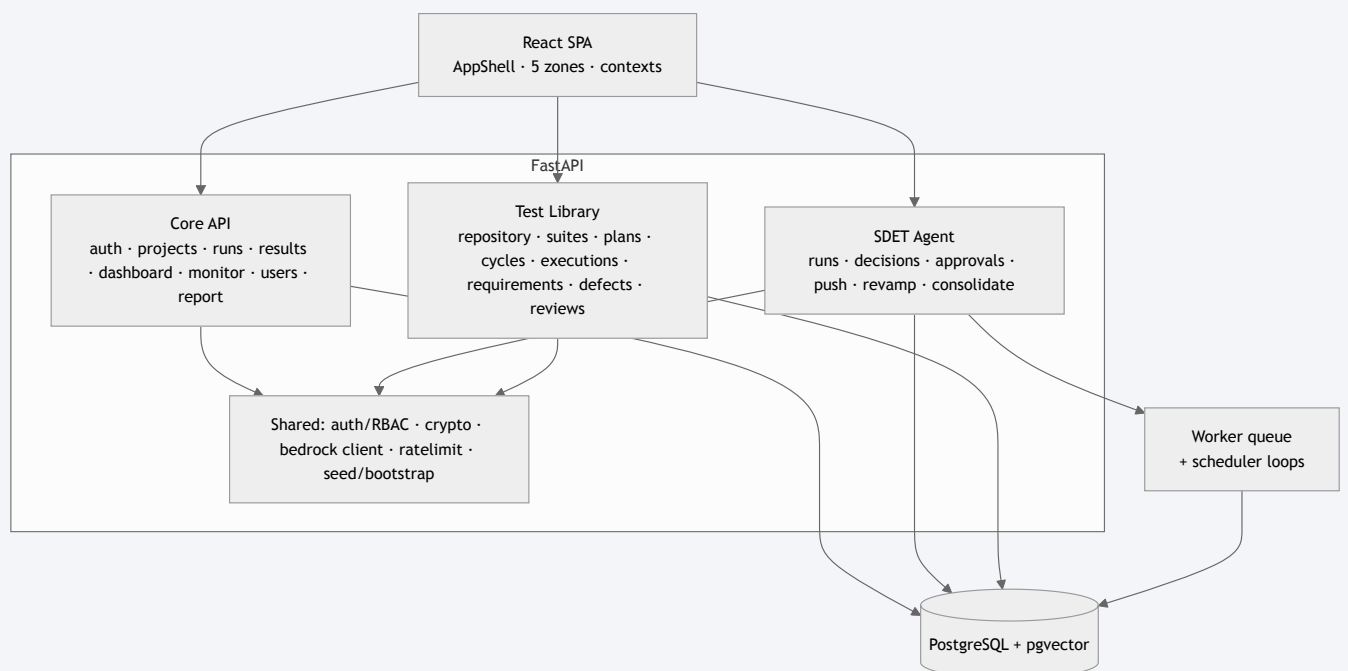
1. System context (who talks to what)



Authentication: password (bcrypt → HS256 JWT) or optional OIDC SSO (off unless configured) — both yield the same app JWT.

TLS is terminated by a reverse proxy in front of Runview (nginx/Traefik); the app itself listens on 127.0.0.1:8080 .

2. Internal components



- **Core** — app/routes.py , services.py , models.py , ingestion parsers, retention, monitor.

- **SDET Agent** — app/sdet/* (agent, retrieval, approvals, worker, scheduler, testsigma_writer, rovo, embeddings...).
- **Test Library** — app/tm/* (services, suites, execution, automation, requirements, defects, reviews, data_sets, templates, io_csv, integrations, audit).
- **Shared** — auth.py, crypto.py (Fernet), bedrock.py, ratelimit.py, seed.py, sdet/bootstrap.py.

3. Data flows

3.1 CI result ingestion

CI → POST /api/projects/{ref}/runs (X-API-Key)
 → API verifies key (bcrypt) → parse JUnit/Mabl/JSON → persist Run + TestResult(+steps/logs)
 → media uploaded separately (≤25 MB) → Attachment + file on disk
 → dashboards/run views read from DB

3.2 SDET Agent run

User → POST /api/sdet-agent/runs (queued in DB)
 Worker → Jira story + PR/dev-status (+ Rovo) → context
 → Titan embedding → pgvector retrieval (7 lenses)
 → Bedrock (Claude) → decisions REUSE/UPDATE/NEW (+ perf analysis)
 → persist sdet_tc_decisions
 User → review → approve (superadmin) → push → TestSigma (create/update)
 → sdet_tc_pushes records before/after snapshot (rollback-able)
 → optionally promote into Test Library as a tm_test_case

3.3 SSO / OIDC login (optional)

User → "Single sign-on" → GET /api/auth/sso/login
 → redirect to IdP (authorization-code + PKCE; state + nonce stored server-side)
 IdP → GET /api/auth/sso/callback?code&state
 → exchange code at token_endpoint → validate id_token via JWKS (issuer, aud, nonce)
 → match user by email (link-existing-only; unknown → rejected)
 → issue normal app JWT → SPA via URL fragment (#sso_token=...)
 (Off by default; superadmin enables/configures it in Settings → Access — stored in AppSetting, client secret encrypted; env OIDC_* is a fallback. Superadmin password break-glass always works.)

3.4 Automation ↔ Test Library mapping

RunView TestResult(classname,name,status) — match by automation_identity (classname::name) → name → case_id → tm_test_cases
 → coverage report (automated / matched / rate)
 → cycle auto-sync sets tm_cycle_cases.exec_status from latest run
 → coverage gaps = automated tests with no matching case

4. Data model (grouped)

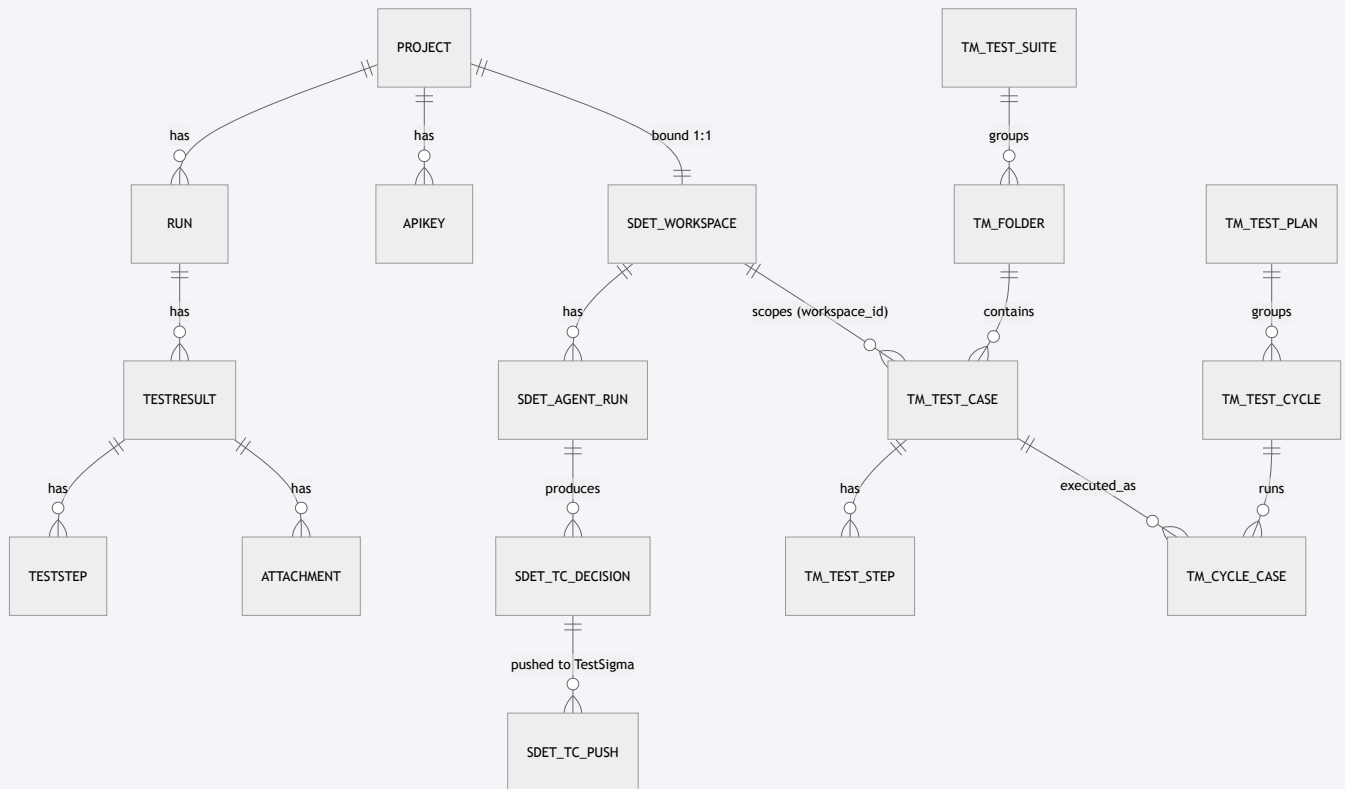
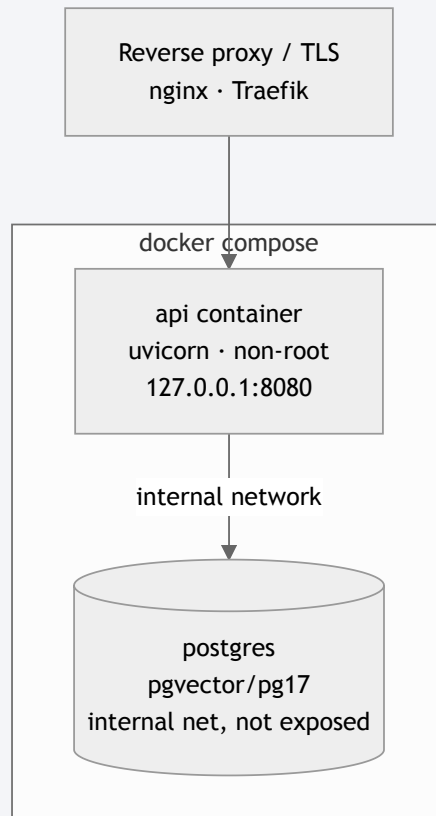


Table groups

- **Core (~12):** project, run, testresult, teststep, testlog, attachment, user (incl. additive SSO columns auth_provider / sso_subject / sso_issuer), apikey, appsetting, dashboardwidget, instancemapping, aiusage.
- **SDET (~20):** sdet_workspaces, sdet_agent_runs, sdet_tc_decisions, sdet_approvals, sdet_tc_pushes, sdet_stories, sdet_pull_requests, sdet_run_events, sdet_testsigma_tcs, sdet_context_docs, kb_product_pages, kb_salesforce_metadata, code_symbols / code_references, historical_defects, integrations, cache_sources, consolidations, dep-graph tables — vector columns use HNSW cosine indexes.
- **Test Library (24):** tm_folders, tm_test_cases, tm_test_steps, tm_test_case_versions, tm_labels, tm_label_assignments, tm_custom_fields (+options/values), tm_settings, tm_test_suites, tm_test_suite_cases, tm_test_plans, tm_test_cycles, tm_cycle_cases, tm_requirements, tm_requirement_links, tm_defects, tm_data_sets, tm_attachments, tm_external_mappings, tm_status/value/field/folder_mappings, tm_audit_events.

5. Deployment topology



- **Build:** multi-stage — Node builds SPA → copied into Python runtime image.
- **Startup:** `init_db` → gated seed → `bootstrap_safe` (Alembic upgrade head + upsert workspaces) → background loops.
- **Migrations:** additive Alembic chain to head **0022**; auto-run when `SDET_AUTO_MIGRATE` enabled.
- **Hardening:** non-root UID 10001, dropped capabilities, `no-new-privileges`, mem/PID limits, internal-only Postgres.

See **SECURITY_AUDIT.md** for controls and gaps, **PRODUCT_DOCUMENTATION.md** for feature detail.